

Part A – Problem Framing & Theory Notes

Red & White Skill Education | Practical Exam | Supervised Learning | Set C

Q1. Why is Accuracy a terrible metric for fraud detection?

Answer:

Fraud datasets are highly imbalanced. For example, 99.83% of transactions may be legitimate while only 0.17% are fraudulent.

If a model predicts every transaction as "Not Fraud", it can achieve approximately 99.83% accuracy.

However, it detects no fraud cases.

Metirc	Result
Accuracy	99.83%
Fraud Precision	0
Fraud Recall	0

Why is accuracy misleading?

The model looks highly accurate because the legitimate class is the overwhelming majority. In reality, all fraud cases are missed.

Therefore, accuracy alone is a terrible metric for evaluating a fraud-detection model.

Confusion Matrix for the all-Not-Fraud model

	Predicted Fraud	Predicted Not Fraud
Actual Fraud	0(TP)	All fraud cases missed (FN)
Actual Legit	0(FP)	All fraud cases correct (TN)

Q2. What is the Precision-Recall tradeoff?

Answer:

Precision measures how many transactions flagged as fraud are truly fraudulent. Recall measures how many actual fraud transactions are successfully detected.

When the classification threshold is lowered from 0.5 to 0.2, more transactions are classified as Fraud. Therefore, Recall generally increases because more fraud cases are detected, while Precision may decrease because more legitimate transactions can be incorrectly flagged as fraud.

Simple 2x2 Cont Confusion Matrix

	Predicted Fraud	Predicted Not Fraud
Actual Fraud	TP	FN
Actual Legit	FP	TN

Threshold	Precision	Recall
0.5	Higher	Lower
0.2	Lower	Higher

Q3. Explain 3 strategies to handle extreme class imbalance.

1. Random Undersampling

Meaning: Removes samples from the majority (legitimate) class to achieve a more balanced dataset.

Pros: Faster training, smaller dataset, and improved class balance.

Cons: Important legitimate examples may be removed, causing loss of useful information.

2. SMOTE Oversampling

Meaning: Creates synthetic samples for the minority (fraud) class to increase its representation.

Pros: Increases minority-class representation and keeps the original majority-class data.

Cons: Synthetic samples may add noise and can increase the risk of overfitting.

3. `class_weight='balanced'`

Meaning: Gives greater importance to the minority class during model training.

Pros: Simple to implement and does not require removing or creating samples.

Cons: May increase false positives and can be less stable when imbalance is extremely severe.

Q4. What is Precision-Recall AUC (PR-AUC)? Why is it preferred over ROC-AUC for extreme imbalance?

Answer: PR-AUC (Precision-Recall Area Under the Curve) is the area under the Precision-Recall curve. It focuses on the performance of the positive/minority class, which is the Fraud class in this problem.

Why is PR-AUC preferred for extreme class imbalance?

- . Fraud cases are rare compared with legitimate transactions.
- . PR-AUC directly reflects Precision and Recall for the fraud class.
- . ROC-AUC can look good even when the model performs poorly on rare fraud cases because the majority legitimate class dominates the overall evaluation.

Q5. In a fraud detection context, which is more costly: a False Positive or a False Negative?

False Positive (FP): A legitimate transaction is incorrectly flagged as fraud.

Specific business example: A customer makes a genuine =50,000 payment, but the fraud system blocks it as suspicious.

Business impact: Customer inconvenience, declined genuine transactions, poor customer experience, and possible loss of trust.

False Negative (FN): A fraudulent transaction is incorrectly classified as legitimate.